Om Thakur

Backend Engineer | Distributed Systems | Scalable APIs

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omthakur.in

PROFESSIONAL SUMMARY

Backend Software Engineer focused on distributed systems, event-driven processing, scalable APIs, and production reliability. Experienced in building asynchronous services, low-latency WebSocket pipelines, secure integrations, and database-backed applications using TypeScript, Python, and PostgreSQL.

EXPERIENCE

Founding Software Engineer
Apr 2026 – Present

Krishora Technologies · Mumbai, India

- Engineered a low-latency Python processing pipeline integrating Pipecat, Exotel WebSockets, model inference, and CUDA-accelerated processing, achieving approximately 700 ms turnaround.
- Built the Node.js/PM2 service router for Sevak AI, utilizing Qdrant and the nomic-embed-text embedding model for semantic retrieval.
- Deployed and maintained self-hosted production infrastructure with Nginx reverse proxying, Cloudflare SSL, PM2 process management, Dograh, and Mailcow.

Software Developer
Aug 2025 – Apr 2026

Cyber Intelligence and Research Lab (CIARL) · Thane, India

- Developed Kotlin/Java Android background services with encrypted SQLite queues to guarantee zero field data loss during intermittent connectivity.
- Engineered React/PostgreSQL applications featuring authentication, fine-grained role-based access control (RBAC), and user management.
- Built a legal-domain RAG pipeline with document ingestion, semantic retrieval, and citation-backed responses; evaluated Matrix-based protocols for secure real-time communication.

Software Development Engineer Intern
Aug 2024 – Feb 2025

Sumeru Technology Solutions Pvt. Ltd. · Bengaluru, India

- Replaced polling-based cron workflows with the Flexbase message-queue service, reducing data synchronization time from 15 minutes to 4 seconds (~99.5% reduction).
- Participated in root cause analysis of a production database outage, identifying payment cron retry bursts that saturated database connections.
- Migrated legacy PHP services to TypeScript and Deno Edge Functions; evaluated notification architectures (Novu) and integrated Supabase Row-Level Security (RLS).

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, Python, SQL, Bash
Backend & Distributed	Node.js, Deno, REST APIs, WebSockets, Message Queues, Webhooks, Asynchronous Pipelines
Databases	PostgreSQL, Supabase, MySQL, Redis, SQLite, Qdrant
Infrastructure & Security	Linux, Docker, Nginx, PM2, Cloudflare SSL, RBAC, Row-Level Security (RLS), Git

SELECTED PROJECTS

Digital Commerce & Entitlement Platform
store.sowebuild.in

Next.js 15 · TypeScript · Supabase · Razorpay

- Implemented dual-verification checkout combining synchronous payment API validation with HMAC-SHA256 webhook verification.
- Encrypted digital assets at rest using AES-256-GCM authenticated encryption with SHA-256 key derivation and randomized IVs.

devpic — Stateless Mathematical SVG API
github.com/MonsterFlick/devpic

TypeScript · Next.js Edge Runtime

- Built an open-source, stateless REST API generating deterministic SVG patterns, geometric meshes, and UI assets via mathematical algorithms.
- Achieved sub-10 ms response times deployed on Next.js Edge Runtime with zero database or persistent storage overhead.

EDUCATION

Bachelor of Science in Information Technology (B.Sc. IT)
Aug 2021 – Apr 2024

Sonubhau Baswant College, University of Mumbai · Mumbai, India

- CGPA: 8.55 / 10 · Graduated with First Rank in College · Led the final-year capstone engineering project team.